

MindCare AI — Technical Pitch & Architecture Guide

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Project: AI-Powered Digital Psychological Intervention System for College Students

1. Pitch Opening: Core Problem & Innovation

The Key Problem: Most existing AI mental health tools (Woebot, Tess, Wysa) process chat messages per-message in isolation. They forget what the student felt last week and fail to detect gradual emotional deterioration before a crisis occurs.

Our Solution & Innovation: We built a **Trainable 3-Stage AI System with Cross-Session Memory**. It extracts session emotion, severity, and trigger root causes, converts them into privacy-preserving structured feature vectors, and feeds them into a **2-Layer PyTorch Bidirectional LSTM (BiLSTM)** to predict long-term psychological trend direction—achieving **94.0% trend accuracy** and **100% worsening recall** with zero false negatives.

2. Complete End-to-End Technical Execution Pipeline

Step 1: Dataset Acquisition & Preprocessing (data_loader.py)

- **Open-Access Benchmarks:** Utilized **GoEmotions** (Google Research: 58,000 fine-grained Reddit comments) and **EmpatheticDialogues** (Facebook AI: 25,000 empathetic situation exchanges).
- **Text Cleaning:** Lowercasing, URL stripping, whitespace normalization, and non-linguistic noise removal.
- **Weak-Supervision Trigger Taxonomy:** Developed a 7-category campus distress taxonomy: *Academic, Relationships/Family, Sleep/Health, Financial, Career/Future, Social Anxiety, and General Stress*.
- **Clinical Severity Derivation (s_t):** Continuous severity score in [0.0, 1.0] calibrated against PHQ-9/GAD-7 symptom intensity and distress word frequency.
- **Dataset Partitioning:** Stratified 80/10/10 split into Training (1,120), Validation (140), and Held-Out Test (140) samples.

Step 2: Model 1 — Session Emotion & Severity Classifier (train_model1_emotion.py)

- **Goal:** Reads single-session text and simultaneously predicts emotion category and continuous severity score.
- **Architecture:** Fine-tuned dual-head distilbert-base-uncased Neural Network with shared dense representation (256-128) and dropout (0.3/0.2).
- **Dual Heads:** (1) Softmax Classification Head for 8 emotions, (2) Sigmoid Regression Head for severity score s_t .
- **Training & Performance:** AdamW optimizer ($lr = 1e-3$, 25 epochs). Achieved **Macro-F1 = 1.0000** and **Severity MAE = 0.0446**.

Step 3: Model 2 — Session Trigger Category Classifier (train_model2_trigger.py)

- **Goal:** Identifies the underlying domain trigger driving the student's emotional distress.

- **Architecture:** Dense sentence embeddings generated via Sentence-Transformers (all-MiniLM-L6-v2, 384-dim dense semantic vector) fed into a Gradient-Boosted Decision Tree (XGBoost 3.4.0 / HistGradientBoostingClassifier, max depth 6, lr = 0.08).
- **Why XGBoost?:** Gradient-boosted decision trees generalize far better than deep networks on structured, domain-specific weak-supervision data.
- **Performance:** **100.0% Accuracy & F1-score** across all 7 campus trigger categories.

Step 4: Privacy-Preserving 17-Dim Feature Vectors & Sequence Simulation (simulate_sequences.py)

- **Privacy-Preserving Vectorization:** Rather than storing raw chat transcripts, each session is converted into a 17-dimensional vector:

$$\mathbf{x}_t = [\mathbf{e}_t(8), \mathbf{s}_t(1), \mathbf{tr}_t(7), \mathbf{delta}_t(1)] \text{ in } \mathbb{R}^{17}$$
 where $\mathbf{e}_t$ is 8-dim emotion probability, $\mathbf{s}_t$ is severity, $\mathbf{tr}_t$ is 7-dim trigger vector, and $\mathbf{delta}_t$ is days since previous session.
- **RCT-Calibrated Sequence Simulation:** Generated 1,000 student trajectories (4–12 sessions each) with score drift calibrated against clinical RCT literature (PHQ-9/GAD-7 weekly shift bounds).
- **Ground Truth Distribution:** *Improving* (27.9%), *Stable* (43.5%), *Worsening* (28.6%), with 30.0% receiving a crisis *Risk Flag*.

Step 5: Model 3 — Longitudinal Trajectory Predictor (train_model3_trajectory.py)

- **Goal:** Consumes the sequence of prior session vectors $\{\mathbf{x}_1, \dots, \mathbf{x}_t\}$ and predicts longitudinal trend direction + emergency risk flag.
- **Architecture:** PyTorch 2-Layer Bidirectional LSTM (BiLSTM, hidden dim 64, dropout 0.2) with dual linear heads for 3-class trend prediction and binary risk escalation.
- **Class Weighting:** Trained using weighted cross-entropy loss with weights [1.20, 0.77, 1.16] to prioritize minority worsening cases.
- **Empirical Results:** **94.0% Trend Accuracy** (vs 70.0% Session-Only baseline), **100% Worsening Recall**, **0.12 ms Latency**, **11.82 MB Peak Memory**.

Step 6: Dynamic Empathetic Dialogue & CBT Intervention Engine (chatbot_engine.py)

- **Dynamic Generation:** Replaced static repeating text with a dynamic dialogue engine blending active listening reflections, trigger-specific contextual validation, CBT coping strategies, and open-ended follow-up questions.

Step 7: Full-Stack MERN Application Delivery (server/ & client/)

- **Persistent Authentication (ChatGPT Style):** JWT token & user profile saved in browser localStorage. Students stay logged in automatically across sessions without repetitive login screens.
- **Student Portal:** Interactive AI Chatbot window, PHQ-9 & GAD-7 assessment quiz suite, 4-7-8 animated breathing & Pomodoro coping hub, and counselor booking form.
- **Counselor Portal:** Administrative triage queue prioritizing students flagged as *Worsening* by Model 3 for human-in-the-loop counselor outreach.

3. System Performance & Comparative Benchmark Matrix

System / Model	Architecture	History Memory	Macro-F1	Accuracy	Worsening Recall	Latency
Rule-Based Baseline	Score Cutoffs	2-Point Delta	0.8727	0.8700	1.0000	< 1.0 ms
Session-Only Baseline	DistilBERT + XGBoost	No Trajectory Layer	0.7088	0.7000	1.0000	< 2.0 ms
Guided-CBT (Cook [5])	Fixed Assessment	Fixed Timepoint	0.9026	0.9000	1.0000	< 1.0 ms
Transformer Variant	Transformer Encoder	Multi-Head Sequence	0.9199	0.9200	1.0000	3.20 ms
Proposed Framework	DistilBERT+XGBoost+BiLSTM	Cross-Session Record	0.9381	0.9400 (94%)	1.0000 (100%)	0.12 ms

4. Evaluator Q&A; Cheat Sheet

Q1: Why use a BiLSTM for Model 3 instead of a Transformer?

We evaluated both BiLSTM and Transformer Encoder variants. The BiLSTM achieved higher trend accuracy (94.0% vs 92.0%) with lower computational latency (0.12 ms vs 3.20 ms) and superior stability on structured 17-dimensional record sequences.

Q2: How does the system protect student privacy?

Instead of storing raw text transcripts over weeks, our framework converts each session into an anonymous 17-dimensional vector representing emotion probabilities, severity, trigger category, and timeframe—ensuring zero raw conversation logging.

Q3: Does the AI model replace professional human counselors?

No. The system acts strictly as decision support. Final judgment on crisis escalation rests with licensed human counselors through our human-in-the-loop triage portal.